

A Research Question:

How can convolutional neural networks (CNNs) improve the detection and classification of Earth-like exoplanets within the habitable zone using light curve data from the Kepler Mission?

Summary:

This research focuses on developing a method for detecting and classifying exoplanets, particularly Earth-like candidates within the habitable zone, using data from the Kepler Mission. The project leverages Artificial Intelligence (AI), specifically Convolutional Neural Networks (CNNs), to automate and enhance the analysis of light curve data, a key source for detecting planetary transits. By employing a novel dual-branch CNN architecture, the method combines global views of the entire light curve with localized, zoomed-in views of potential transit events to improve the detection accuracy of subtle signals often missed by traditional approaches.

The aim is to not only identify planetary candidates but also to distinguish false positives from genuine exoplanets, reducing the need for extensive manual review. This approach addresses key challenges in exoplanet detection, such as working with noisy and incomplete datasets, while contributing to the broader scientific objective of understanding planetary system formation and the potential for life beyond Earth. By automating this process, the research aligns with and extends current advancements in AI applications in astrophysics, offering a scalable and efficient solution for processing large astronomical datasets.

Methodology:

Data Collection and Preprocessing:

- Utilize light curve datasets from the NASA Kepler Mission, which include orbital period, transit depth, and stellar properties.
- Clean and preprocess the dataset to create global (entire light curve) and localized (zoomed-in transit) views for model training.
- Normalize numerical features and encode categorical labels to prepare the data for machine learning models.

Model Development:

- Develop a dual-branch convolutional neural network (CNN) architecture:
 - One branch analyzes global light curve patterns to capture overall brightness variations.
 - The other branch focuses on localized transit events to detect fine-grained features.
 - Integrate auxiliary stellar features through a separate branch of dense layers.
- Train and refine the hybrid CNN model to classify exoplanet candidates, false positives, and confirmed exoplanets.

Training and Validation:

- Split the dataset into training, validation, and test sets.
- Train the models using GPU acceleration for faster processing.
- Validate the model's performance using metrics like accuracy, precision, recall, F1-score, and confusion matrices.

Implementation and Evaluation:

- Implement the trained model on unseen datasets (e.g., TESS data) to evaluate its robustness.
- Continuously optimize the model based on performance and feedback to improve its generalizability.

Resources Needed:

- Lab Computer: Necessary for processing large light curve datasets and training models with dual-branch CNN architecture effectively. This includes running experiments with high batch sizes and multiple epochs.
- Google Colab Pro: Essential for accelerating the computationally intensive deep learning tasks, such as training convolutional neural networks and processing high-dimensional astronomical data.
- Professor's Guidance: Valuable for expert advice and feedback to refine the methodology, address challenges, and ensure the research aligns with astrophysical objectives.

Annotated Bibliography:

1. https://www.researchgate.net/profile/Saima-Khan-22/publication/383429169_A_Review_of_Exoplanet_Detection_Processes_using_Artificial_Intelligence_and_Neural_Networks/links/66cd453cc2eaa500231addb5/A-Review-of-Exoplanet-Detection-Processes-Using-Artificial-Intelligence-and-Neural-Networks.pdf

This paper reviews the use of artificial intelligence (AI), especially convolutional neural networks (CNNs), in exoplanet detection. It focuses on how AI improves the accuracy and speed of analyzing light curve data, making it easier to find exoplanets while reducing false positives. The paper summarizes various methods researchers have developed and highlights the potential of AI to handle the challenges of processing large astronomical datasets. By leveraging advanced techniques, the study demonstrates how AI can broaden our understanding of planetary systems and accelerate discoveries. However, I felt like the paper could provide more detailed evaluations of different AI approaches to offer a clearer comparison of their effectiveness.

2. https://aissmsioitresearch.com/wp-content/uploads/2024/01/IJTEAMS-ISS2_03.pdf

This paper explores the use of artificial intelligence (AI), particularly convolutional neural networks (CNNs), in exoplanet detection. It highlights how AI has enhanced the efficiency and accuracy of analyzing light curves, which helps detect exoplanets with

reduced false positives. The study provides an overview of how deep learning models, like CNNs, are applied to large datasets generated by space missions such as Kepler and TESS. The research effectively demonstrates AI's transformative role in exoplanet discovery, but I felt like it could benefit from more detailed comparisons between traditional and AI-based detection methods to better evaluate their respective strengths and limitations.

3. <https://norma.ncirl.ie/4400/1/nandhavarmanjeevarathinam.pdf>

This study explores the use of deep learning, specifically the VGG16 convolutional neural network (CNN), to classify light curves from the Kepler mission for detecting exoplanets. The research applies novel preprocessing techniques, including recurrence plots and image augmentation, to enhance pattern recognition in the noisy and sparse light curve data. The study achieves a classification accuracy of 94% using recurrence plots, outperforming traditional methods like Random Forest. While effective, the model suffers from overfitting, indicating a need for future improvements. I felt like this research paper offered more information on the subject matter and it breaks it down well.